

Perception for a Robotic Mealtime Assistant

Problem and applications

This project proposal is for developing a camera-based system for safer, more respectful feeding robots. It identifies when a person is "Ready" to be fed (facing forward, mouth open, silent) using facial recognition and an image classifier for head pose and lip movement. Unsure cases default to "Not-Ready" for safety.

In this project, I treat feeding readiness as a temporal classification problem on short video clips rather than on isolated frames. Each clip contains at least one transition from Not Ready to Ready, for example, the resident turning their head back to the spoon or stopping a conversation and opening their mouth. The goal is to estimate a clean sequence of Ready or Not Ready labels over time that updates as the state changes, which is what a mealtime robot will need in order to decide when to present food and when to wait.

Relationship to course content

The project is a focused application of perception topics: image sensing and quality considerations; geometric reasoning with landmarks; probabilistic decision making; and available CNNs for classification. It exercises classical vision (landmarks, PnP head-pose estimation) alongside a compact learned component (a mouth-crop classifier) and emphasizes measured evaluation with clear safety-relevant operating points, mirroring the course's emphasis on principled perception pipelines and metrics.

The initial results in this report focus on a compact pipeline that uses MediaPipe keypoints and simple temporal statistics to classify Ready versus Not Ready windows in a clip. This follows the course's emphasis on interpretable feature extraction and clear evaluation metrics for perception systems. In the second phase, I will add a small CNN-based sequence model on cropped mouth regions as a learned comparison, as suggested in the project feedback, and compare it directly with the engineered baseline.

Solution method

Instead of training from scratch and making CNN architecture, the reasonable approach will not to train big neural network but to use already developed architecture and build upon them for my application. Each video frame goes through **MediaPipe Face Landmarker/Face Mesh** to find reliable key points around the eyes, nose, and lips.

For the initial phase I implemented a fully engineered baseline that operates on upper body and hand keypoints extracted from RGB video. Each frame is processed by MediaPipe Holistic, which returns 33 body landmarks and 21 landmarks per hand. From these landmarks, I constructed a small set of feeding-related features per frame: the distance from each index fingertip to a rough mouth centre, the vertical position of each wrist relative to the shoulders, the

elbow joint angle on each side, and approximate hand speeds over time. These features are smoothed with a short median filter to reduce jitter.

I collected manual segment labels for each video in terms of continuous time intervals that are Ready or Not Ready. These segment labels are converted to frame-level labels by matching the time stamps, and then aggregated into sliding windows of about one second of video with a half-second stride. For each window, I computed simple statistics of the engineered features, such as means, standard deviations, minima, maxima, and slopes. Each window receives a Ready or Not Ready label by majority vote over its frames.

The current baseline classifier is a Random Forest trained on these window-level features. It outputs a probability of Ready for each window, which can later be turned into a stable time series of labels using thresholds and a short dwell time to reduce flicker. The same pipeline will be used on a larger dataset with multiple participants and subject held-out splits, and will serve as the main baseline for comparison with a small CNN sequence model on mouth crops in the final report.

Datasets and ground truth

Public face and speech datasets are used to tune and test the MediaPipe and OpenCV pipeline before validating on a small custom volunteer set. Each personal short video clip is manually labeled as READY, or NOT READY based on face stability, mouth, and head direction.

Some relevant Models & Dataset:

- https://ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker/python
- https://mediapipe.readthedocs.io/en/latest/solutions/face_mesh.html
- <https://github.com/google-ai-edge/mediapipe/wiki/MediaPipe-Face-Mesh>
- https://docs.opencv.org/4.x/d5/d1f/calib3d_solvePnP.html
- https://docs.opencv.org/4.x/d7/d53/tutorial_py_pose.html
- https://docs.opencv.org/4.x/dd/d6a/classcv_1_1KalmanFilter.html
- <https://www.kaggle.com/datasets/davidvazquezcic/yawn-dataset>
- <https://www.kaggle.com/datasets/hoangtung719/drowsiness-dataset>
- <https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html>
- EatSense. (2023). Retrieved November 15, 2025, from Ed.ac.uk website: <https://groups.inf.ed.ac.uk/vision/DATASETS/EATSENSE/>
- Pexels. (2025). Retrieved November 15, 2025, from Pexels.com website: <https://www.pexels.com/search/videos/chewing/>
- Pexels. (2025). Retrieved November 15, 2025, from Pexels.com website: <https://www.pexels.com/search/videos/eating/>

For the initial results, I used a small pilot dataset of short face and upper body clips recorded with a stationary RGB camera at 720p resolution. Each clip contains at least one transition from Not Ready to Ready, for example, the participant looking away, then returning and opening their mouth. For each pilot clip, I manually defined time segments in seconds that are Ready or Not Ready, and stored them in a simple CSV file. A script converts these segment-level labels into frame-level labels by matching each frame time to the appropriate segment, which then drives the window-level labels used for training and evaluation.

By the final report, I plan to scale this to approximately ten to twelve participants, each contributing five to six clips of eight to twelve seconds. The clips will cover a mix of scenarios such as talking then pausing to eat, turning the head away then back, or short occlusions near the mouth. I will create train, validation, and test splits at the subject level so that the model is always evaluated on people it did not see during training, which addresses the generalization concerns raised in the project feedback. The same labeling protocol is used for all clips, with at least one Not Ready to Ready transition annotated per clip.

Implementation status

The full engineered baseline pipeline is now implemented and tested on the pilot clip. There are scripts to:

1. Extract MediaPipe Holistic keypoints and save per-frame pose and hand landmarks.
2. Convert keypoints into feeding-related engineered features such as hand-to-mouth distance, wrist height, elbow angle, and hand speed.
3. Align manual segment labels to frames and produce frame-level Ready and Not Ready labels.
4. Build one-second sliding windows with a half-second stride and compute window-level feature statistics.
5. Train a Random Forest classifier on the window features, split into train and test windows, and report basic metrics.

Running this pipeline on the pilot clip produces a window-level dataset with both classes present and shows that the engineered features behave in the expected way. For example, the minimum hand-to-mouth distance drops and wrist height rises during Ready segments, where the hand approaches the mouth, while speeds are higher during transitions.

Goals/Timeline for semester

The overall goal remains to deliver a real-time Ready or Not Ready detector that runs on short face clips and can be integrated into a mealtime robot. After the initial results milestone, the project is divided into completed work and upcoming work.

Completed so far

- Set up the repository, virtual environment, and configuration files so all scripts run in a single project layout.
- Implemented the engineered baseline pipeline from raw MP4 videos to labeled window features using the scripts `make_label_templates.py`, `align_segments_to_frames.py`, `extract_keypoints.py`, `build_features.py`, `build_window_features.py`, a small split script, `train_baseline.py`, and `plot_look.py`.
- Processed the pilot clip end-to-end and confirmed that both Ready and Not Ready windows are present, the baseline model achieves non-trivial accuracy on held-out

windows, and the quick look plots match expected behavior, such as smaller hand-to-mouth distances during Ready periods.

Next steps

- Collect additional eating clips with at least one Not Ready to Ready transition, label segments, run the full pipeline for each clip, and grow the window-level dataset.
- Refine data splits to operate at the video and subject level, retrain the baseline on the multi-video dataset, and report accuracy and F1 scores on held-out participants.
- Implement a small CNN-based sequence model on mouth crops as a comparison to the engineered baseline, then prepare a short demo video with overlaid predictions and integrate all results into the final report.

Quantitative and qualitative results

```
PS C:\AER1515\Project\Code\Perception-for-a-Robotic-Mealtime-Assistant> python scripts/train_baseline_model.py
Loading train_data.csv and test_data.csv
Raw label values:
Label
ready      5692
not_ready  3122
not_ready    30
Name: count, dtype: int64
Dropping rows with other labels:
Label
not_ready  30
Name: count, dtype: int64
Using feature columns: ['lh_to_mouth', 'rh_to_mouth', 'lh_speed', 'rh_speed', 'lwrist_rel_shoulder_y', 'rwrist_rel_shoulder_y', 'left_elbow_angle_deg', 'right_elbow_angle_deg']
Class distribution after filtering:
1    5692
0    3122
Name: count, dtype: int64
Raw label values:
Label
ready      575
not_ready  375
Name: count, dtype: int64
Using feature columns: ['lh_to_mouth', 'rh_to_mouth', 'lh_speed', 'rh_speed', 'lwrist_rel_shoulder_y', 'rwrist_rel_shoulder_y', 'left_elbow_angle_deg', 'right_elbow_angle_deg']
Class distribution after filtering:
1     575
0     375
Name: count, dtype: int64
Training done.

Test accuracy: 0.682

Confusion matrix (rows true, cols predicted):
[[165 210]
 [ 92 483]]

Classification report:

```

	precision	recall	f1-score	support
not_ready	0.64	0.44	0.52	375
ready	0.70	0.84	0.76	575
accuracy			0.68	950
macro avg	0.67	0.64	0.64	950
weighted avg	0.68	0.68	0.67	950

```

Saved model to C:\AER1515\Project\Code\Perception-for-a-Robotic-Mealtime-Assistant\results\features\baseline_rf.joblib
Saved importances to C:\AER1515\Project\Code\Perception-for-a-Robotic-Mealtime-Assistant\results\features\baseline_rf_feature_importance.csv
Saved confusion matrix plot to C:\AER1515\Project\Code\Perception-for-a-Robotic-Mealtime-Assistant\results\features\confusion_matrix.png
Saved feature importance plot to C:\AER1515\Project\Code\Perception-for-a-Robotic-Mealtime-Assistant\results\features\feature_importance.png

```

